

Problem Set 1

1. Sleep.c

```
xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ sleep 10
$
```

Shell correctly waited 10 seconds before moving to the next instruction (\$). I don't know why test failed.

2. Sixfive.c

```
xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ echo "5 100 18 6 xv6 hello 30-25" > sixfive.txt
$ sixfive sixfive.txt
5
100
18
6
6
30
25
$
```

3. Memdump.c

```
$ cat > input.txt
deadc0de
$ memdump hhcccc < input.txt
25956
25697
c
0
d
e
$
```

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4. Find.c

```
xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ echo > b
$ mkdir a
$ echo > a/b
$ mkdir a/aa
$ echo > a/aa/b
$ find . b
./b
./a/b
./a/aa/b
$ |
```

5. Exec.c (find modification but made a separate program named exec.c):

```
$
$
$ echo > b
$ mkdir a
$ echo > a/b
$ mkdir a/aa
$ echo > a/aa/b
$ exec . b -exec echo hi
hi ./b
hi ./a/b
hi ./a/aa/b
$ QEMU: Terminated
marium@DESKTOP-0K07MJA:~/xv6-labs-2025$
```

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6. Make grade:

Exec modification of find fails the test because I did not modify find.c instead created a new program exec.c so that I don't mess up the working find.c and keep separate clear concepts.

```
o kernel/sleeplock.o kernel/file.o kernel/pipe.o kernel/exec.o kernel/sysfile.o kernel/kernelv
rt.o kernel/console.o kernel/printf.o kernel/uart.o kernel/spinlock.o
riscv64-unknown-elf-ld: warning: kernel/kernel has a LOAD segment with RWX permissions
riscv64-unknown-elf-objdump -S kernel/kernel > kernel/kernel.asm
riscv64-unknown-elf-objdump -t kernel/kernel | sed '1;/SYMBOL TABLE/d; s/ .*/ / ; /' > ke
make[1]: Leaving directory '/home/marium/xv6-labs-2025'
== Test sleep, no arguments ==
$ make qemu-gdb
sleep, no arguments: OK (3.0s)
== Test sleep, returns ==
$ make qemu-gdb
sleep, returns: OK (0.7s)
== Test sleep, makes syscall ==
$ make qemu-gdb
sleep, makes syscall: OK (1.1s)
== Test sixfive_test ==
$ make qemu-gdb
sixfive_test: OK (1.1s)
== Test sixfive_readme ==
$ make qemu-gdb
sixfive_readme: OK (0.9s)
(Old xv6.out.sixfive_readme failure log removed)
== Test sixfive_all ==
$ make qemu-gdb
sixfive_all: FAIL (1.1s)
...
init: starting sh
$ sixfive sixfive.txt README
GOOD 5
GOOD 100
GOOD 10
GOOD 6
$ qemu-system-riscv64: terminating on signal 15 from pid 5162 (make)
MISSING '^1810'
MISSING '^1810'
QEMU output saved to xv6.out.sixfive_readme
== Test memdump, examples ==
$ make qemu-gdb
```

```
== Test memdump, examples ==
$ make qemu-gdb
memdump, examples: OK (1.0s)
== Test memdump, format ii, S, p ==
$ make qemu-gdb
memdump, format ii, S, p: OK (0.9s)
== Test find, in current directory ==
$ make qemu-gdb
find, in current directory: OK (1.2s)
== Test find, in sub-directory ==
$ make qemu-gdb
find, in sub-directory: OK (0.9s)
== Test find, recursive ==
$ make qemu-gdb
find, recursive: OK (1.3s)
== Test exec ==
$ make qemu-gdb
exec: FAIL (0.6s)
...
hart 2 starting
init: starting sh
$ find . wc -exec echo
Usage: find <path> <filename>
$ qemu-system-riscv64: terminating on signal 15 from pid 5354 (make)
MISSING '^./wc$'
QEMU output saved to xv6.out.find_exec
== Test exec, multiple args ==
$ make qemu-gdb
exec, multiple args: FAIL (1.1s)
Number of appearances of 'hello'
got:
1
expected:
5
QEMU output saved to xv6.out.find_exec_args
== Test exec, recursive find ==
$ make qemu-gdb
exec, recursive find: FAIL (1.2s)
```

```
== Test exec, recursive find ==
$ make qemu-gdb
exec, recursive find: FAIL (1.2s)
Number of appearances of 'hello'
got:
0
expected:
3
QEMU output saved to xv6.out.test_find_sh
== Test time ==
time: OK
Score: 91/131
make: *** [Makefile:360: grade] Error 1
marium@DESKTOP-0K07MJA:~/xv6-labs-2025$ |
```